Advanced Generative AI

Innomatics Research Labs — Course Syllabus

NASSCOM FutureSkills Prime Certified Advanced Generative AI Program. The official syllabus covers AI/GenAI foundations, LLMs, prompting, OpenAI, Gemini, LLaMA, Hugging Face, LangChain, RAG and Stable Diffusion.

Source: https://www.innomatics.in/advanced-generative-ai-course/

Module 1: Introduction to Artificial Intelligence

- Introduction to AI
- AI vs ML vs DL
- Supervised, unsupervised and reinforcement learning
- Core differences between ML and DL
- ML/DL project lifecycle

Module 2: Introduction to Generative AI

- Introduction to Generative AI
- Overview of generative AI technologies
- Applications and industry case studies

Module 3: Getting Started with Large Language Models

- Introduction to LLMs
- History of NLP
- RNN, LSTM and GRU
- Encoder-decoder models

Module 4: Prompt Engineering and Working with LLM

- Prompt engineering fundamentals
- LLMs with prompt engineering
- GPT models and GPT-3/GPT-4 concepts
- Training and applications of GPT models
- Generating text and code
- LLM case study

Module 5: Working with OpenAI API

- OpenAI introduction and APIs
- API authentication and setup
- GPT-3/GPT-4 text generation exercises
- DALL-E and image generation
- Text-to-image hands-on project

Module 6: Working with Google Gemini Gen API

- Getting started with Gemini
- Obtaining an API key

- Gemini API features
- Exploring Gemini models
- Model selection and initialization
- Gemini-powered chatbot project

Module 7: Working with Meta's LLaMA API

- LLaMA introduction and comparison with GPT
- Capabilities and architecture
- Model sizes and variants
- Environment setup
- Text generation with LLaMA

Module 8: Working with Hugging Face Ecosystem

- Hugging Face ecosystem and Transformers
- Models and tokenizers
- Trainer API
- Hugging Face integration with web applications
- Hands-on project

Module 9: Building GenAI Apps using LangChain

- LangChain introduction
- Core components and purpose
- Setup and dependencies
- Development configuration
- Building applications with LangChain

Module 10: Introduction to RAG

- RAG fundamentals
- Building RAG applications
- LLMs in depth
- Fine-tuning LLMs
- Training/fine-tuning hands-on

Module 11: Stable Diffusion by Stability AI

- Stable Diffusion introduction
- Diffusion model fundamentals
- Image generation applications
- Prompt-based image modification
- Seeds, prompts and steps
- Stable Diffusion tools
- Fine-tuning on custom datasets
- Advanced prompt engineering
- DreamBooth and personalization

- Diffusers library
- Web application integration
- Advanced Stable Diffusion techniques

Prepared from the syllabus/curriculum published on the official Innomatics Research Labs website. Website content can change; use the source URL above as the authoritative current reference.